

Subject Description Form

Subject Code	COMP4122
Subject Title	Game Design and Development
Credit Value	3
Level	4
Pre-requisite / Co-requisite / Exclusion	Nil
Objectives	The objectives of this subject are to: 1. provide students with a broad view of the nature of digital games and entertainment; 2. equip students with the knowledge and skills in the analysis, design and development of digital games and, by extension, interactive applications such as extended reality apps; 3. equip students with the knowledge and skills to broadly explore and utilize different algorithms, techniques and tools for game development; 4. equip students with the knowledge, skills and aesthetic sense to manipulate a real-time rendering pipeline and to develop interactive visual effects; 5. equip students with knowledge in game AI, multi-user games and networking for games; and, 6. nurture students' aesthetic sense in game design and guide students to study and evaluate the social impact of digital games.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional/academic knowledge and skills</u> (a) get familiar with the history, evolution and recent development in digital games; (b) grasp and analyse the overall hardware and software architecture of typical 2D/3D digital games; (c) have a deep understanding of different practical analysis, design and implementation techniques that apply to the development of digital games across different platforms; (d) identify important issues of game design, testing, and playability principles; (e) comprehend and grasp the game development and programming aspects, such as event loops, execution threads, rendering, animation, terrain and background representation, polygonal models, texturing, collision detection and physically-based modelling, game AI, and multi-user games and networking;

	(f) assess and appreciate the social impact of computer games; <i>Attributes for all-roundedness</i> (g) work in groups to design and develop a digital game; (h) communicate effectively and present, both in terms of an oral presentation and a written report, a game concept and the result of developing a digital game; and, (i) comprehend and appreciate the broader perspectives of digital entertainment, and social impact of games.								
Subject Synopsis/ Indicative Syllabus	<table><tr><th>Topic</th></tr><tr><td>1. Introduction and game history. Early origins of video games and digital game design and programming; arcade games; some popular home game systems; game design methodology and principles; and, current development in digital game industry.</td></tr><tr><td>2. Game software, hardware and platforms. Web-based; console-based (e.g., PlayStation, Xbox, Switch); PC-based, handheld games; mobile; and, wireless game platforms.</td></tr><tr><td>3. Game design principles. Game psychology and game sociology; human characteristics as information processors and players in gaming environment; cognitive principles for game design; and, game design techniques and case studies.</td></tr><tr><td>4. Programming and implementation techniques in digital games. 3D models; geometry; rotation, and quaternion interpolation; camera control; texturing; terrain; lighting; level of details (LOD); collision detection; shading; and, non-photorealistic rendering.</td></tr><tr><td>5. Game behaviour and game AI. Acceleration; motion analysis; game physics; path finding; chasing and evading; pattern movement; flocking; obstacle avoidance; swarming; learning the opponent behaviours; search and heuristics in games; finite state machines; and, decision trees.</td></tr><tr><td>6. Languages and tools in game development. C#, C, Unity, Unreal and other authoring tools for various components in games, such as terrain generators and node-based editors.</td></tr><tr><td>7. Social impacts of computer games. Social and cultural characteristics in digital games; digital entertainment software rating; future of games.</td></tr></table>	Topic	1. Introduction and game history. Early origins of video games and digital game design and programming; arcade games; some popular home game systems; game design methodology and principles; and, current development in digital game industry.	2. Game software, hardware and platforms. Web-based; console-based (e.g., PlayStation, Xbox, Switch); PC-based, handheld games; mobile; and, wireless game platforms.	3. Game design principles. Game psychology and game sociology; human characteristics as information processors and players in gaming environment; cognitive principles for game design; and, game design techniques and case studies.	4. Programming and implementation techniques in digital games. 3D models; geometry; rotation, and quaternion interpolation; camera control; texturing; terrain; lighting; level of details (LOD); collision detection; shading; and, non-photorealistic rendering.	5. Game behaviour and game AI. Acceleration; motion analysis; game physics; path finding; chasing and evading; pattern movement; flocking; obstacle avoidance; swarming; learning the opponent behaviours; search and heuristics in games; finite state machines; and, decision trees.	6. Languages and tools in game development. C#, C, Unity, Unreal and other authoring tools for various components in games, such as terrain generators and node-based editors.	7. Social impacts of computer games. Social and cultural characteristics in digital games; digital entertainment software rating; future of games.
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Teaching/ Learning Methodology	During the lectures, students will come across the common concepts, methods, and issues in digital game analysis, design and development, and will be supplemented by exercises in games evaluation, and development. During the labs / tutorials, students will have the opportunity to practice, apply, and present what they have learned, and form in groups to develop a computer game which forms a major output of this subject.								

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed								
			a	b	c	d	e	f	g	h	i
	Continuous Assessment										
	1. Project	45%	✓	✓	✓	✓	✓	✓	✓	✓	
	2. Mid-term Programming Exam	10%			✓		✓				
	Examination										
	3. Final Exam	45%	✓	✓	✓	✓	✓	✓			
	Total	100%									
	The course is designed to equip students with the knowledge and skillset of a game developer. Strong practical elements, supported by a theoretical foundation, are components to train the design and development capabilities of students. Students need to complete a major game development project involving concept design, tools/techniques analysis and implementation in an appropriate setting and platform. The game is expected to be a playable product that demonstrates its key ideas. The purpose of the project is to train students’ ability in design and development, and to enrich their portfolio. A mid-term programming exam will assess students’ ability to quickly develop a mini game; this will help develop students' ability to quickly produce prototypes to experiment with game ideas. Students also need to get familiar with some successful/popular games by playing them so as to understand the characteristics and limitations as a game player and game designer, and to broadly analyse and critique digital games to train their design mentality.										
	Student Study Effort Expected	Class contact:									
▪ Lecture							26 Hrs.				
▪ Tutorial/Lab							13 Hrs.				
Other student study effort:											
▪ Coursework and Project							80 Hrs.				
Total student study effort							119 Hrs.				

Reading List and References	1. Rick Parent, <i>Computer Animation: Algorithms and Techniques</i>, 3rd Edition, 2012. 2. Jesse Schell, <i>The Art of Game Design: A Book of Lenses</i>, 3rd Edition, 2019. 3. Jason Gregory, <i>Game Engine Architecture</i>, 3rd Edition, 2018. 4. Computer Animation and Virtual Worlds. 5. IEEE Transactions on Games. 6. Tomas Akenine-Möller, Eric Haines, Naty Hoffman, Angelo Pesce, Michał Iwanicki, and Sébastien Hillaire, <i>Real-Time Rendering</i>, 4th Edition
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